

**CSES003 : EXTERNAL EXPERT INPUT –
DATA STRUCTURE**

Project Report

Plant Disease Recognition System

Submitted in partial fulfillment of the requirements for the award of degree of

B.Tech. (Computer Science & Engineering)

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Student Declaration

I, Faraz Ahmad Khan, Registration Number 12313142 & Shubham Swaraj, Registration Number 12315878 hereby declare that the project entitled "Plant Disease Recognition System" is our original work and has been carried out by us under the prescribed guidelines of Lovely Professional University. This project is submitted in partial fulfillment of the requirements for the award of the degree of Bachelor of Technology (B.Tech.) in Computer Science and Engineering.

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We also extend our gratitude to the creators of the Plant Village Dataset available on Kaggle, which provided the 87,000+ annotated leaf images used to train our model across 38 disease categories.

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Table of Content

Chapter 1: Introduction.....	5
1.1 Importance of Plant Disease Detection.....	5
1.2 Motivation for the Project.....	5
Chapter 2: Project Overview	6
2.1 Dataset Description	6
2.2 Data Cleaning and Transformation	6
Chapter 3: Problem Statement.....	7
3.1 Objectives of the Project.....	7
Chapter 4: System Architecture & Implementation	8
4.1 Model Architecture (CNN).....	8
4.2 Flask Web Application	8
4.3 Disease Prediction Flow	8
4.4 Frontend Interface Design	9
4.5 Disease Information Output	9
Chapter 5: Results and Discussion	10
5.1 Key Observations	10
5.2 Future Scope	11
Chapter 6: Conclusion.....	12
References	13

Chapter 1: Introduction

Agriculture is the backbone of India's economy, supporting the livelihoods of over 140 million farmers across the country. Among the most significant threats to agricultural productivity are plant diseases, which silently destroy 20 to 40 percent of global crop yields every year. Diseases caused by fungi, bacteria, and viruses spread rapidly through fields, often going undetected until an entire harvest is lost.

Early and accurate identification of plant diseases is the single most powerful intervention available to farmers. However, traditional diagnosis relies on physical inspection by agricultural experts or laboratory testing — both of which are inaccessible to the majority of small and marginal farmers in rural areas.

This project, titled "Plant Disease Recognition System", addresses this critical gap by developing an AI-powered web application that enables instant plant disease detection from leaf photographs. By leveraging deep learning and computer vision, the system classifies plant diseases and provides actionable treatment recommendations directly from a smartphone browser.

1.1 Importance of Plant Disease Detection

- Early identification of diseased crops prevents disease spread and saves entire harvests from loss.
- Accurate diagnosis enables targeted treatment, avoiding wasteful and incorrect pesticide use.
- Reduces dependency on expensive agricultural experts and laboratory testing.
- Accessible via smartphone, making expert-level diagnosis available to farmers in remote areas.
- Contributes to food security by minimizing yield losses across major crop types.

1.2 Motivation for the Project

The primary motivation behind this project is to apply deep learning and web technologies to solve a real-world agricultural problem. Despite the availability of large image datasets and powerful machine learning frameworks, plant disease diagnosis tools remain inaccessible to the farmers who need them most.

This project aims to bridge the gap between cutting-edge AI research and practical field application. By training a Convolutional Neural Network (CNN) on the Plant Village dataset and deploying it through a Flask web application, the system delivers instant disease predictions with cause and treatment information — making advanced agricultural intelligence available to anyone with a smartphone.

Chapter 2: Project Overview

The Plant Disease Recognition System is designed as an end-to-end agricultural intelligence application integrating:

- A deep learning CNN model for plant disease classification.
- A Flask-based web application for user interaction and image upload.
- A structured JSON knowledge base for disease cause and treatment information.
- A professional web interface with animated backgrounds and result display.

The system supports real-time prediction and informational output, enabling farmers and agricultural workers to obtain diagnosis and treatment advice within seconds of uploading a leaf photograph.

2.1 Dataset Description

The dataset used in this project is the Plant Village Dataset, sourced from Kaggle and originally created by Penn State University. It is one of the most widely used open-source datasets for agricultural AI research.

- Total Images: 87,000+ annotated leaf photographs
- Disease Classes: 38 categories across 14 crop types
- Image Type: RGB photographs under controlled lighting conditions

Crops Covered:

- Apple, Blueberry, Cherry, Corn, Grape, Orange, Peach
- Pepper (Bell), Potato, Raspberry, Soybean, Squash, Strawberry, Tomato

Each image is labelled with the plant species and its disease condition (or healthy status), providing a rich supervised learning dataset for training the classification model.

2.2 Data Cleaning and Transformation

- Image Resizing: All images resized to 160x160 pixels to match the model input layer.
- Normalization: Pixel values converted to NumPy arrays using Keras utilities.
- Format Conversion: Images converted to NumPy arrays for TensorFlow compatibility.
- Train/Validation Split: Dataset divided into training and validation sets for unbiased model evaluation.

Chapter 3: Problem Statement

Plant diseases are one of the most devastating threats to global food security. Farmers, especially in developing countries, lack timely access to expert diagnosis tools. By the time a disease is visually identified and an expert is consulted, it may have already spread across an entire field — resulting in complete crop failure and financial ruin for smallholder farmers.

Traditional plant disease identification relies on:

- Manual inspection by trained agronomists — expensive and geographically limited.
- Laboratory testing — time-consuming and unavailable in rural regions.
- Visual comparison with disease handbooks — error-prone and impractical in the field.

This project addresses these challenges by developing a centralized, intelligent, and accessible plant disease recognition system that integrates a deep learning classification model with a web-based interface — enabling instant, accurate diagnosis from a simple leaf photograph.

3.1 Objectives of the Project

- Develop a Convolutional Neural Network (CNN) to accurately classify plant diseases from leaf images across 38 disease categories.
- Train the model on the Plant Village dataset containing 87,000+ labelled leaf images spanning 14 crop types.
- Deploy the trained model as a Flask web application accessible through a standard browser on any device.
- Provide structured output including the disease name, its biological cause, and recommended treatment steps for each prediction.
- Design a professional, mobile-friendly interface with image upload, drag-and-drop support, and animated visual feedback.
- Identify healthy plants correctly and distinguish them from the 38 disease classes to avoid false alarms.
- Enable real-time prediction delivering results within seconds of image upload.
- Make the system accessible to non-technical users including farmers and agricultural field workers.
- Support early intervention by flagging disease presence at the earliest visible leaf stage.
- Demonstrate the practical application of deep learning, computer vision, and web deployment in the agricultural domain.

Chapter 4: System Architecture & Implementation

4.1 Model Architecture (CNN)

The core of the system is a Convolutional Neural Network (CNN) built using TensorFlow and Keras. CNNs are specifically designed to process image data by detecting spatial patterns at multiple levels of abstraction — from edges and textures at lower layers to complex disease patterns at deeper layers.

- Input Layer: Accepts 160x160 RGB images (160, 160, 3 tensor shape).
- Convolutional Layers: Multiple Conv2D layers extract local features such as spots, lesions, and discoloration patterns.
- Pooling Layers: MaxPooling2D layers reduce spatial dimensions while retaining dominant features.
- Dropout Layers: Applied to prevent overfitting during training, improving generalization.
- Flatten Layer: Converts 2D feature maps into a 1D vector for classification.
- Dense Layers: Fully connected layers learn the mapping between extracted features and disease classes.
- Output Layer: A softmax layer with 39 neurons outputs probability scores for each of the 38 disease classes plus background.

4.2 Flask Web Application

Flask is a lightweight Python web framework used to serve the application. It handles all communication between the user's browser and the AI model.

Flask uses Jinja2 templating to dynamically inject prediction results — disease name, cause, and treatment — directly into the HTML page, displaying the result on the same page as the upload form without a redirect.

4.3 Disease Prediction Flow

The complete prediction pipeline follows these steps when a user uploads an image:

- Step 1 — Image Upload: The user selects or drags a JPEG or PNG leaf image into the circular upload panel on the homepage.
- Step 2 — File Validation: The application checks the file extension and rejects non-image uploads with an error message.
- Step 3 — Secure Save: A UUID-based filename is generated and the file is saved to the uploadimages/ directory.
- Step 4 — Preprocessing: The image is loaded and resized to 160x160 pixels using `tf.keras.utils.load_img()`, then converted to a NumPy array.
- Step 5 — Model Inference: The preprocessed image array is passed to the loaded CNN model which outputs a 39-value probability vector.

- Step 6 — Argmax Classification: `numpy.argmax()` identifies the class index with the highest probability score.
- Step 7 — JSON Lookup: The index retrieves the corresponding disease entry from `plant_disease.json` containing name, cause, and cure.
- Step 8 — Result Rendering: Flask renders `home.html` with prediction data and image path, displaying the diagnosis card to the user.

4.4 Frontend Interface Design

The web interface was designed to be professional, visually engaging, and accessible on mobile devices. Key design features include:

- **Animated Background:** Four agricultural background images cycle every 5 seconds using CSS transitions and JavaScript.
- **Circular Upload Panel:** A three-ring concentric circle design with green glow effects housing the upload form. Clicking anywhere on the circle triggers the file picker.
- **Drag and Drop Support:** Users can drag leaf images directly onto the circle with visual feedback on hover.
- **Live Thumbnail Preview:** After file selection, a circular thumbnail of the chosen image appears below the upload panel.
- **Branding Panel:** The right side displays the application name, three feature highlights, and a description of the system.
- **Result Card:** After prediction, a dark-themed card appears showing the uploaded image, disease name, cause, and treatment.
- **About Section:** Explains the agricultural problem, problem vs solution comparison, how it works in 3 steps, and a closing mission statement.
- **Contact Section:** Cards listing key agricultural institutions with website links and Google Maps directions.

4.5 Disease Information Output

The `plant_disease.json` file serves as the knowledge base. It contains 39 entries — one per class. Each entry contains three fields:

Field	Description
name	The full disease label e.g. Tomato — Late Blight
cause	The biological or environmental cause of the disease
cure	Recommended treatment steps and preventive measures

This structured output ensures the system not only identifies the disease but equips the farmer with actionable information to treat and prevent further crop damage.

Chapter 5: Results and Discussion

This chapter presents the outcomes obtained from the implementation of the Plant Disease Recognition System and discusses the insights derived from model evaluation and system testing. The results demonstrate the effectiveness of using a Convolutional Neural Network trained on the PlantVillage dataset for real-time agricultural disease classification.

The CNN model trained on 87,000+ leaf images successfully learned to distinguish between 38 disease classes and healthy plant conditions. Predictions are delivered within seconds of image upload, making the system practical for real-world field use.

Disease Classification Results:

- The model correctly classifies fungal diseases such as Apple Scab, Tomato Late Blight, and Potato Early Blight with high confidence.
- Bacterial diseases including Tomato Bacterial Spot and Peach Bacterial Spot are distinguished accurately from similar-looking fungal conditions.
- Viral diseases such as Tomato Yellow Leaf Curl Virus and Tomato Mosaic Virus are correctly identified despite subtle visual signatures.
- Healthy plant conditions are correctly classified, preventing false alarms that could lead to unnecessary pesticide application.

Crop-wise Performance:

- Tomato: Highest number of disease classes (9 categories) — all classified correctly in testing.
- Potato: Early blight and late blight distinguished accurately despite similar visual symptoms.
- Grape: Four disease categories including Esca (Black Measles) correctly identified.
- Corn: Cercospora leaf spot distinguished from Northern Leaf Blight successfully.

5.1 Key Observations

- Deep learning models provide highly effective plant disease classification from leaf images without laboratory analysis.
- The PlantVillage dataset produces a highly generalizable model for common crop diseases.
- Flask enables seamless deployment of the TensorFlow model as a web service accessible from any browser.
- JSON-based knowledge output enhances practical utility by providing actionable cause and treatment alongside the classification.
- Drag-and-drop interface and instant results make the system genuinely usable by non-technical farmers in field conditions.

5.2 Future Scope

Future enhancements to this project may include:

- Integration of a larger and more diverse dataset including images captured under real field conditions — varying lighting, angles, and backgrounds — to improve model generalization beyond controlled lab photographs.
- Multi-disease detection in a single image to identify cases where a plant suffers from more than one disease simultaneously, using object detection architectures such as YOLO or Faster R-CNN.
- Severity level estimation — extending the model to not only classify the disease but also estimate its severity (mild, moderate, severe) to guide treatment urgency and dosage recommendations.
- Mobile application deployment using TensorFlow Lite, enabling the model to run directly on a smartphone without requiring a server connection — ideal for areas with limited internet access.
- Regional language support — translating disease names, causes, and treatments into Hindi, Punjabi, Telugu, and other Indian languages to improve accessibility for farmers not proficient in English.
- Integration with government agricultural databases to automatically connect farmers with subsidized treatment resources and expert helplines upon diagnosis.
- Real-time drone or IoT sensor integration to enable large-scale field scanning, detecting disease outbreaks across entire farms before they become visible to the human eye.
- Transfer learning using pre-trained architectures such as ResNet50, VGG16, or EfficientNet to improve accuracy and reduce training time on expanded datasets.

Chapter 6: Conclusion

The Plant Disease Recognition System successfully demonstrates the effective application of deep learning, computer vision, and web development technologies in solving a critical agricultural challenge. By leveraging the PlantVillage dataset and a Convolutional Neural Network trained on 87,000+ leaf images, the project delivers a practical, accurate, and accessible tool for plant disease identification.

The integration of the TensorFlow/Keras classification model with a Flask web application enables real-time disease prediction from a simple leaf photograph — providing farmers and agricultural workers with instant diagnosis, cause information, and treatment recommendations without requiring any technical knowledge or access to laboratory facilities.

The professionally designed web interface with animated backgrounds, circular upload panel, drag-and-drop support, and structured result cards ensures that the system is not only technically capable but also genuinely usable by its intended audience — farmers in the field.

The About section contextualizes the problem at a national scale — illustrating how 140 million Indian farmers and 20–40% of global crop yields are affected by plant diseases annually — reinforcing the real-world significance of this system beyond its academic purpose.

The Contact Resources section connects users with established agricultural institutions including Krishi Vigyan Kendra Delhi, G. B. Pant University, Agrio, and IARI — ensuring users who require expert assistance know where to turn.

Overall, this project illustrates how modern AI tools can be meaningfully applied to improve agricultural outcomes, reduce crop losses, and support food security — while remaining accessible, affordable, and practical for deployment in real-world farming communities. Although designed for educational purposes, it provides a strong foundation for future expansion into a production-grade agricultural intelligence platform.

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